VS

Python vs Ruby

ARRAY SHUFFLE PROBLEM

CSE 451: Concepts of Programming Languages

The Problem

Input

Given a string **s** and an integer array **indices** of the same length.

```
s = "art"
indices = [1, 0, 2]
```

Output

Shuffle string so character at position **i** moves to **indices[i]**.

```
"rat"
```

'a' (idx 0) → idx 1
'r' (idx 1) → idx 0
't' (idx 2) → idx 2

Algorithm Step-by-Step

- **Initialize:** Create an empty result array of size *N*.
- **Iterate:** Loop through the input string with index *i*.
- **Place:** Assign result[indices[i]] = s[i].
- **Join:** Convert the result array back into a string.

Complexity: **O(N)** Time **|** **O(N)** Space

Python

The "Batteries Included" Language

- **Created:** 1991 by Guido van Rossum.
- **Philosophy:** "There should be one-- and preferably only one --obvious way to do it."
- **Key Traits:** Dynamic typing, huge ecosystem (pip), explicit syntax.
- **Uses:** AI/ML, Data Science, Web (Django), Scripting.

Python Implementation

Key Features

- enumerate(s) gives us both index and character.
- List initialization: [''] * len(s) is idiomatic.
- ''.join() is the standard way to build strings.

```python
def shuffle_string(s, indices):
    # Create empty list of size N
    result = [''] * len(s)

    for i, char in enumerate(s):
        # Map current char to target index
        result[indices[i]] = char

    return ''.join(result)
```

Ruby

"Developer Happiness"

- **Created:** 1995 by Yukihiro "Matz" Matsumoto.
- **Philosophy:** "Optimize for programmer happiness."
- **Key Traits:** Pure OOP (everything is an object), flexible syntax (TIMTOWTDI), blocks.
- **Uses:** Web Dev (Ruby on Rails), DevOps (Puppet/Chef).

Ruby Implementation

Key Features

- Array.new creates the structure.
- each_char.with_index chains enumerators elegantly.
- **Block Syntax:** do |char, i| ... end is central to Ruby.
- Implicit return (last line returns).

```ruby
def shuffle_string(s, indices)
  # Initialize array with defaults
  result = Array.new(s.length, '')

  # Iterate with index
  s.each_char.with_index do |char, i|
    result[indices[i]] = char
  end

  # Implicit return
  result.join
end
```

Feature Comparison

Feature	Python	Ruby
Type System	Dynamic, Strong	Dynamic, Duck Typing
Array Init	[''] * len(s)	Array.new(n, '')
Iteration	enumerate(s)	s.each_char.with_index
Block Syntax	Indentation (whitespace)	do/end or { }
String Join	''.join(list)	array.join

Challenges & Learnings

Environment Setup

Installing Ruby on Windows required the RubyInstaller and DevKit configuration to manage native extensions.

Syntax Curve

Adapting to Ruby's block syntax and method chaining (e.g., .with_index) was a shift from Python's direct style.

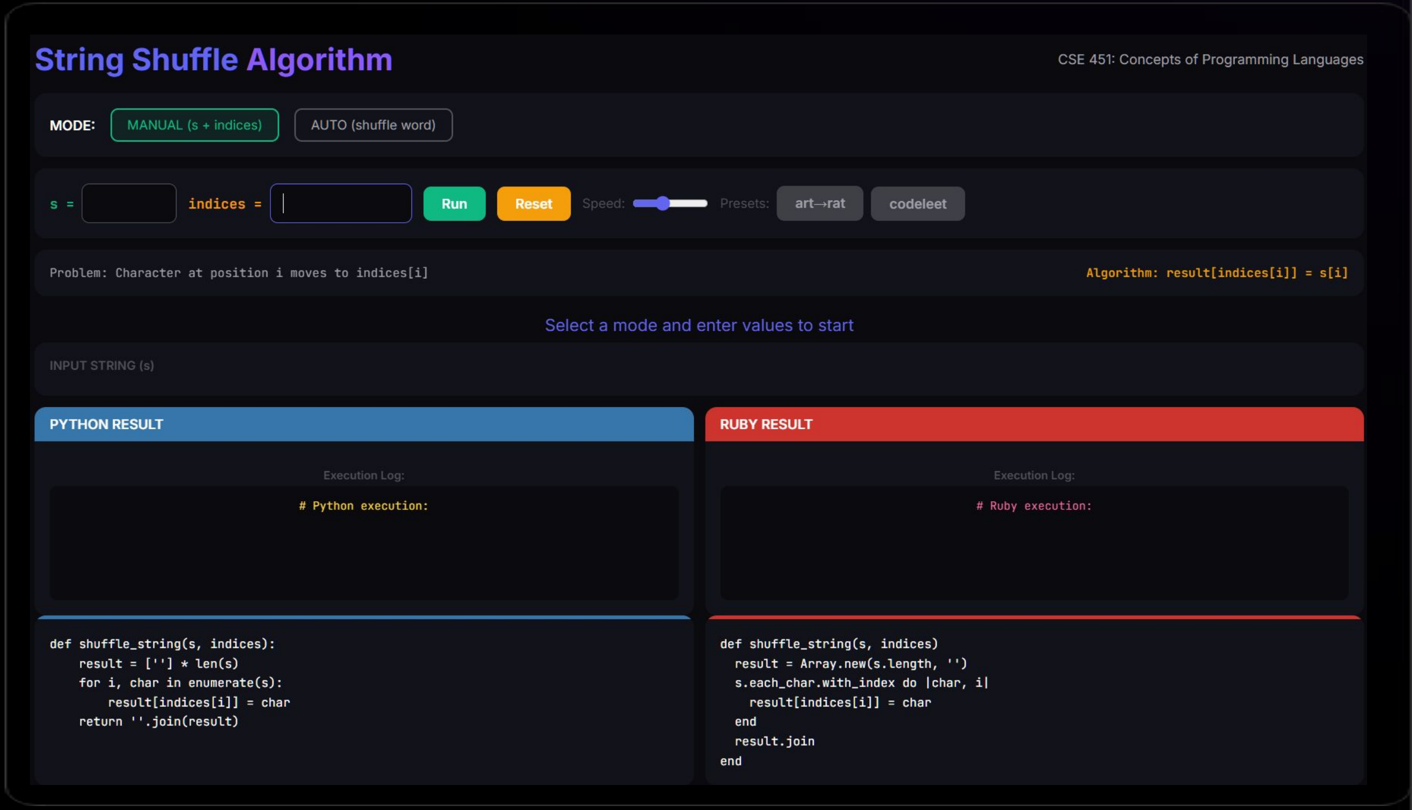
Verification

Ensuring both implementations produced identical output for edge cases (empty strings, single characters).

Live Demo

Try the App Yourself !

Scan to see the algorithm in action.



https://mvvhmxd.github.io/Python-vs-Ruby

Key Takeaway

"Understanding the **concepts** (arrays, indices, complexity) matters more than the **syntax**."

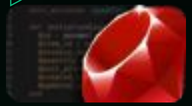
Questions?

Thank you for listening.

Image Sources

- 
  https://img.freepik.com/premium-vector/set-abstract-colorful-spheres-from-points-lines-dark-background-network-connection-structure-big-data-visualization-vector-illustration_634443-1804.jpg
  Source: www.freepik.com

- 
  https://4kwallpapers.com/images/wallpapers/python-logo-dark-2880x1800-16094.png
  Source: 4kwallpapers.com

- 
  https://i.ytimg.com/vi/vXV6Ahqw0Sk/hq720.jpg?sqp=-oaymwEhCK4FEIIDSFryq4qpAxMIARUAAAAAGAElAADIQj0AgKJD&rs=AOn4CLCYT3rxW-go1QJhrdcJ36MFtfaaLA
  Source: www.youtube.com

- 
  https://res.cloudinary.com/practicaldev/image/fetch/s--rUTLfE4---/c_limit%2Cf_auto%2Cfl_progressive%2Cq_auto%2Cw_880/https://dev-to-uploads.s3.amazonaws.com/i/jrxbe5fpuuk6mgchng1m.png
  Source: dev.to